

NICK MARKS

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EXPERIENCE

Relevant skills: Python, C++, Java, CMake, Docker, Cloud, Bash

SWE - AWS Networking

DC Lifecycle and Tooling

📅 09/2025 - Present 📍 Sydney, AUS

- Led design and build of a new automated intake system responding to fiber optic planning signals. This system saved 2+ months of engineering time per year and increased speed average intake creation speed by 50%.

SWE - AWS Relational Database Services

RDS Control Plane Tooling

📅 01/2025 - 09/2025 📍 Sydney, AUS

- Led development of a control plane alarming service with suppression of alarms during high fault-volume instances. This saved RDS/Aurora teams 20+ hours of operational work upon release of fault injection service features to external customers.
- Enhanced a deadline monitoring service receiving 1+ million workflows per hour with "smart" deadlines. This cut or extended workflow timeouts based on historical data to reduce operational load for teams by 32% each week.

PROJECTS

Autonomous Navigation through Offline Deep Learning

📅 07/2023 - 06/2024 📍 Monash University

- Researched applications of offline deep learning techniques to autonomously navigate a Penguin Pi bot around varying complexities of road tracks.
- Crafted pipeline for training and deploying a model trained via Implicit Q-Learning, using tools like regression analysis and grid-search to optimise trained experts.

Rocket Guidance Systems at [Purdue Orbital](#)

📅 01/2023 - 05/2023 📍 West Lafayette, IN

- Produced control systems on Purdue Orbital's guidance and navigation team for rockets launched by the team in the middle of 2023 as part of Project Icarus.
- Crafted state models and Simulink implementations of rocket attitude control and implemented a real-time operating system to interface between the control system, onboard sensors and diagnostic systems.

EDUCATION

BSc Computer Science and Pure Mathematics
BE Electrical and Computer Systems Engineering

Monash University (WAM: 87.8%, GPA: 3.88/4.00)

📅 03/2020 - 06/2024 📍 Melbourne, VIC

- *Relevant coursework (BSc):* Advanced Algorithms, Data Structures, Operating Systems, AI, Computer Architecture, Linear Algebra, Advanced Calculus, Modelling and Simulation, Group Theory, Real & Functional Analysis, Graph Theory.
- *Relevant coursework (BE):* Embedded Systems, Digital Systems and Logic, Deep Learning, Probability Modelling, FPGA/ASIC Design, Electrical Circuits, Signals and Systems, Information Theory, Networking, Energy/Power Systems, Web-app Development, Fluid Mechanics.

Semester Abroad

Purdue University (WAM: 92.1%, GPA: 4.00/4.00)

📅 01/2023 - 05/2023 📍 West Lafayette, IN

- *Relevant coursework:* Integrated Circuit Design, Control Systems, Electromagnetics, Artificial Intelligence (AI) and Machine Learning.

COMPETITIONS & RESEARCH

ISC21 Student Cluster Competition (2nd prize)

📅 11/2021 📍 Remote

- Optimised builds of scientific applications written in Fortran, such as XCompact3D.
- Bench-marked builds simulations running on distributed systems, improving parallel performance with OpenMP.

A-HUG Cloud Computing Hackathon

📅 07/2021 📍 Remote

- Ported scientific and AI applications to an ARM-based architecture and bench-marked the relevant differences in performance between the ARM architecture and the native one.

Robot-assisted Drone Navigation at [Monash Deep-Neuron](#)

📅 08/2022 - 11/2022 📍 Monash University

- Developed a behavioral cloning-based approach to train a AI-assisted drone navigation system capable of flying point-to-point within simulation.
- Evaluated the model performance in comparison to a reinforcement-learning approach on a dataset generated by human participants in a small-scale trial (~30 participants).

CO2 Conversion Project Lead

[Monash Carbon Capture and Conversion](#)

📅 08/2021 - 12/2023 📍 Melbourne, VIC

- Founded a large-scale research project on the electrochemical reduction of CO2 into fuels in a novel, catalysed flow cell environment.
- Our team received US\$250,000 as a top 23 university finalist worldwide in the [XPRIZE Carbon Removal Competition](#) .

AWARDS & ACHIEVEMENTS

- Dean's honour roll for Faculties of Engineering and Science (2020-2022).
- School Captain of Scotch College, conducted post-graduation in a full-time capacity (2019).
- Graduated high school with ATAR of 99.95 (top 0.05% of country). Co-Dux of School, Dux of Literature (2018).
- Recipient of [SAGSE](#)  Scholarship for German Language Exchange (2018).

LANGUAGES

English
German
Spanish

